

# Differential Equations and Slope Fields

## ANSWER KEY



### Verifying solutions to differential equations

- 1 Verify that  $y = Ce^{3x}$  is a solution to the differential equation  $\frac{dy}{dx} = 3y$  for any constant  $C$ .  
 $\frac{dy}{dx} = 3Ce^{3x} = 3(Ce^{3x}) = 3y$ . This confirms  $y = Ce^{3x}$  satisfies the equation for any  $C$ .
- 2 Verify that  $y = x^2 + C$  is a general solution to  $\frac{dy}{dx} = 2x$ , and find the particular solution satisfying  $y(1) = 5$ .  
 $\frac{dy}{dx} = 2x$  ✓. Using  $y(1) = 5$ :  $1 + C = 5 \Rightarrow C = 4$ . Particular solution:  $y = x^2 + 4$ .

### Solving separable differential equations

- 3 Solve the separable differential equation  $\frac{dy}{dx} = \frac{x}{y}$ , given  $y(0) = 3$ .  
 $y dy = x dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$ . Using  $y(0) = 3$ :  $9 = 0 + 2C \Rightarrow 2C = 9$ . Solution:  $y^2 = x^2 + 9$ , i.e.  $y = \sqrt{x^2 + 9}$  (positive root, since  $y(0) = 3 > 0$ ).
- 4 Solve  $\frac{dy}{dx} = 2xy$  given  $y(0) = 1$ .  
 $\frac{1}{y} dy = 2x dx \Rightarrow \ln|y| = x^2 + C$ . Using  $y(0) = 1$ :  $\ln 1 = 0 + C \Rightarrow C = 0$ . So  $\ln y = x^2 \Rightarrow y = e^{x^2}$ .
- 5 Solve  $\frac{dy}{dx} = \frac{\cos x}{y^2}$  given  $y(0) = 2$ .  
 $y^2 dy = \cos x dx \Rightarrow \frac{y^3}{3} = \sin x + C$ . Using  $y(0) = 2$ :  $\frac{8}{3} = 0 + C \Rightarrow C = \frac{8}{3}$ . So  $y^3 = 3\sin x + 8$ , i.e.  $y = \sqrt[3]{3\sin x + 8}$ .

### Slope fields

- 6 For the differential equation  $\frac{dy}{dx} = x - y$ , find the slope at the points  $(0, 0)$ ,  $(1, 0)$ , and  $(1, 1)$ , as would be drawn on a slope field.  
At  $(0, 0)$ : slope =  $0 - 0 = 0$ . At  $(1, 0)$ : slope =  $1 - 0 = 1$ . At  $(1, 1)$ : slope =  $1 - 1 = 0$ .
- 7 Explain how a slope field for  $\frac{dy}{dx} = ky$  (exponential growth/decay) would look different for  $k > 0$  versus  $k < 0$ .  
For  $k > 0$ : slopes are positive above the  $x$ -axis and negative below, steepening as  $|y|$  increases — curves grow away from  $y = 0$  (unstable equilibrium, exponential growth). For  $k < 0$ : slopes point toward  $y = 0$  from both sides, flattening as  $|y| \rightarrow 0$  — curves decay toward  $y = 0$  (stable equilibrium, exponential decay).

### Exponential growth and decay models

- 8 A bacteria culture grows according to  $\frac{dP}{dt} = 0.2P$ , with  $P(0) = 500$ . Find  $P(t)$  and the population after 10 hours.  
Solution form:  $P(t) = P_0 e^{kt} = 500e^{0.2t}$ .  $P(10) = 500e^2 \approx 500(7.389) \approx 3694$  bacteria.
- 9 A radioactive substance decays according to  $\frac{dA}{dt} = -0.05A$ . If  $A(0) = 100$  grams, find how long it takes to decay to 50 grams (the half-life).  
 $A(t) = 100e^{-0.05t}$ . Set  $50 = 100e^{-0.05t}$ :  $0.5 = e^{-0.05t}$ ,  $\ln(0.5) = -0.05t$ ,  $t = \frac{\ln(0.5)}{-0.05} = \frac{-0.693}{-0.05} \approx 13.86$  years.

## Matching slope fields to differential equations

- 10 A slope field shows horizontal segments (slope 0) along the entire  $x$ -axis and along the entire  $y$ -axis, with slopes elsewhere resembling the product  $xy$ . Identify the differential equation this slope field represents.  
 $\frac{dy}{dx} = xy$  — slope is 0 whenever  $x = 0$  or  $y = 0$ , matching the described field.

## Newton's Law of Cooling

- 11 An object cools according to  $\frac{dT}{dt} = -k(T - 70)$ , where  $70^\circ\text{F}$  is room temperature. If  $T(0) = 200^\circ\text{F}$  and  $k = 0.1$ , find  $T(t)$ .  
Let  $y = T - 70$ :  $\frac{dy}{dt} = -0.1y \Rightarrow y = y_0 e^{-0.1t}$ .  $y_0 = 200 - 70 = 130$ . So  $T(t) = 70 + 130e^{-0.1t}$ .

## Visualizing an exponential growth solution

- 12 The graph shows the bacteria population  $P(t) = 500e^{0.2t}$  from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using  $\ln 2$ .  
The graph crosses  $P = 1000$  at approximately  $t \approx 3.5$ . Algebraically:  $1000 = 500e^{0.2t}$  gives  $2 = e^{0.2t}$ , so  $t = \frac{\ln 2}{0.2} \approx 3.47$ , matching the graphical estimate.

## A logistic-style differential equation

- 13 A population satisfies  $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$ . Find the equilibrium solutions (where  $\frac{dP}{dt} = 0$ ), and explain which one is stable for a population starting near  $P = 10$ .  
Setting  $\frac{dP}{dt} = 0$ :  $P = 0$  or  $P = 200$ . Starting near  $P = 10$  (close to 0 but positive), the population will grow toward the stable equilibrium  $P = 200$  (the carrying capacity), since  $P = 0$  is unstable for any positive starting population.

## Additional separable equations

- 14 Solve  $\frac{dy}{dx} = \frac{y}{x}$  given  $y(1) = 6$ .

$\frac{1}{y}dy = \frac{1}{x}dx \Rightarrow \ln|y| = \ln|x| + C \Rightarrow y = Ax$  for a constant  $A$ . Using  $y(1) = 6$ :  $A = 6$ :  $y = 6x$ .

- 15 Solve  $\frac{dy}{dx} = \frac{1}{y}$  given  $y(0) = 4$ .

$y dy = dx \Rightarrow \frac{y^2}{2} = x + C$ . Using  $y(0) = 4$ :  $8 = 0 + C \Rightarrow C = 8$ . So  $y^2 = 2x + 16$ , i.e.  $y = \sqrt{2x + 16}$  (positive root, since  $y(0) = 4 > 0$ ).

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## A mixing problem modeled by a differential equation

- 16 A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If  $y(t)$  is the amount of salt (kg) at time  $t$ , the differential equation is  $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$ . Find the equilibrium value of  $y$ , and explain what it represents physically. Setting  $\frac{dy}{dt} = 0$ :  $6 = \frac{y}{25}$  gives  $y = 150$  kg. This represents the long-run salt content: eventually the rate of salt entering (6 kg/min from the inflow) balances the rate leaving, keeping the amount constant at 150 kg.